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CLOUD NATIVE GEOSPATIAL: REALIZING THE DIGITAL EARTH VISION

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ABSTRACT

Since Al Gore originally pitched his vision for a Digital Earth in 1999, we've made significant progress towards its realization. Earth observation (EO) data is no longer shared in a raw form, it is processed to agreed analysis ready data specifications and made available at a global scale. The cloud optimized geotiff has emerged as the predominate data format for storing EO data and the spatio-temporal asset catalog specification has been broadly adopted, providing a common method of describing and discovering available data. This combination of technologies and capabilities represents a new paradigm for working with the vast collections of data now accessible on the public cloud and is referred to as cloud native geospatial.

This paper covers background concepts relating to cloud native geospatial, then charts the history of some Digital Earth programs and places them in context with the new availability of global analysis ready data collections. It then provides some examples of how this new capability can be used to enable more people to access more data more easily.

Index Terms **—** Data Analysis, Societal Engagement and Impacts, Big Data, Cloud Computing, Analysis Ready Data, Cloud Native Geospatial.

1. INTRODUCTION

Twenty-five years ago, Al Gore spoke about his vision for a Digital Earth [1], where the full range of data about our planet and our history would be at our fingertips. Recent advances in the availability of Committee on Earth Observation Satellites (CEOS) Analysis Ready Data (ARD) [2] as well as the emergence of the cloud native geospatial paradigm [3] together provide the foundations for the infrastructure that we can use to continue the realization of the Digital Earth vision.

In this paper, we first cover some background concepts, before discussing the history of Digital Earth platforms, which charts a journey from lifting data out of offline archives and creating foundational ARD products through to delivering these data in open access cloud-native geospatial data collections. This new wave of interoperable data collections empowers people from all over the world to use Earth observation data more easily. As Al Gore said:

> *"In the long run, we should seek to put the full range of data about our planet and our history at our fingertips." [1]*

2. BACKGROUND

2.1. CEOS Analysis Ready Data for Land

In optical remote sensing, ARD is created by applying standardized processing and systematic error removal to raw data. Figure 1 shows a data processing workflow starting from raw data, through to Level 1 (terrain corrected) to Level 2 (spectrally calibrated), which is analysis ready.

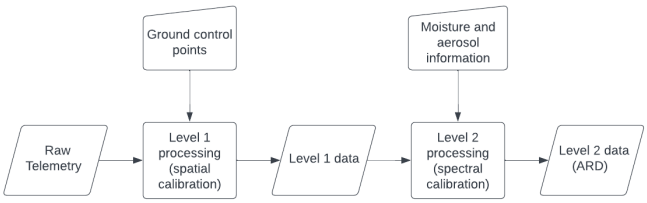


Figure 1: A simplified view of a data processing workflow, progressing from raw telemetry from the satellite through to analysis ready data. Diagram modified from [4].

The CEOS ARD for Land (CARD4L) product family specifications (PFS) were established to develop agreement on what a general-purpose ARD product for surface reflectance is [4]. These specifications provide an important baseline of data processing levels, metadata, and ancillary data products, which CEOS certifies, enabling people to assess whether the data products are appropriate for their use case.

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CARD4L helped define and standardize the processing levels for ARD and the specifications are now used by organisations such as the European Space Agency (ESA), Geoscience Australia (GA), and the United States Geological Survey (USGS) as a benchmark [2]. CARD4L has since evolved into CEOS-ARD, expanding beyond just land, to include additional specifications such as Aquatic Reflectance.

2.2. Cloud Native Geospatial

Cloud native geospatial isn't just a suite of technologies, it's more a combination of technology with ways of working and of thinking about work that make full use of cloud computing. It's a change in thinking, from the lift-and-shift practices of the early cloud, where an existing workload was "lifted" to the cloud as is, to using as-a-service and other abstractions of computation infrastructure. An example of a cloud native technology is Kubernetes, which can be used to run workloads without specifically managing storage, memory, or compute resources. Below we discuss several cloud native geospatial tools and how they work together to represent a new way of working with geospatial data on the cloud.

2.2.1. Cloud Optimized GeoTIFF

In 2015, Planet, in collaboration with Amazon Web Services, uploaded Landsat level 1 data onto the cloud in the Cloud Optimized GeoTIFF (COG) format accompanied by simple metadata [5]. The COG is a backwards compatible extension of the GeoTIFF geospatial image format, and its innovation is in allowing for small parts of the data to be read over the Internet in a simple way.

The COG and other cloud native geospatial formats (such as Zarr and Geoparquet) have proven fundamental to the modern way of working with large volumes of data, as they enable people to stream subsets of data directly across the network, rather than needing to download data to a local file system first. Any slowdown in terms of network latency when reading from the cloud as compared with from a local disk is generally made up in massive gains in parallelized read bandwidth through horizontal scaling made available by the cloud.

2.2.2. Spatio-Temporal Asset Catalog

Spatio-Temporal Asset Catalog (STAC) is a specification for describing geospatial information in a way that helps both humans and machines discover data. It is not a geospatial metadata document but instead a geospatial dataset-level document, which describes what a dataset contains without needing to open it. When working with Landsat or Sentinel-2 data, a STAC item references a scene as a specific location at a specific time and describes the multiple files (often COGs) that relate to the red, green, blue, and other bands as separate assets attached to the item.

In this way, many STAC items can be used together to develop a plan for loading and processing some or all the data over many scenes. STAC also defines an API standard, so in effect, there's an index of a vast data collection, managed by the data custodian, which is publicly exposed. For Landsat, USGS' STAC API can be used to find and load Landsat scenes from anywhere on the Earth's land surface over the 40 years of Landsat's history, and it takes seconds.

2.3. Interoperability

Together, COGs and STAC mean that an organization or individual who is managing a collection of data can expose that data in a cloud native data format and with data-driven metadata that is compatible with what is used by others. It's already possible to mix data from NASA, USGS, Microsoft's Planetary Computer (MSPC) and Element-84's Sentinel-2 collection using common Python libraries and a few lines of code. Figure 2 shows a simple example of searching for, loading and visualizing Sentinel-2 data over a region of interest in Tasmania, Australia. It's very simple to then do a second, similar query and combine another data source, such as Landsat.

```
from pystac_client import Client
from odc.stac import load

client = Client.open("https://earth-search.aws.element84.com/v1")
collection = "sentinel-2-l2a"
tas_bbox = [146.5, -43.6, 146.7, -43.4]
search = client.search(collections=[collection], bbox=tas_bbox, datetime="2023-12")

data = load(search.items(), bbox=tas_bbox, groupby="solar_day", chunks={})
data[["red", "green", "blue"]].isel(time=2).to_array().plot.imshow(robust=True)
```

16.6s

<matplotlib.image.AxesImage at 0x2bfd7dd50>

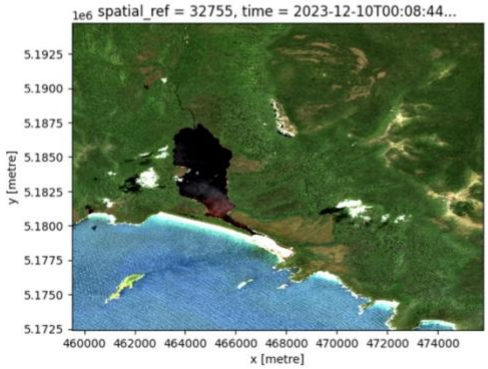


Figure 2, a snippet of Python code and resulting image showing how to load and visualize Sentinel-2 data from COGs using a STAC API.

This availability of global, interoperable, cloud native data with a common discovery API in STAC is core to the cloud native geospatial paradigm and has the potential to be as impactful as the original opening of the Landsat archive in 2008 [6].

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3. SELECTED HISTORY OF DIGITAL EARTHS

3.1. Digital Earth Australia

Digital Earth Australia (DE Australia) was launched by Geoscience Australia in 2017 [6] but has a much longer history. Australia manages a downlink station for Landsat, and historically as part of this station's work, they stored a large volume of Landsat data on offline tape archives. The Unlocking the Landsat Archive project commenced in 2011 [7] and aimed to lift the offline data to modern computer disks, making it accessible, but also to process the data to Level 2 (ARD). This production of standardized ARD takes a long time as it is a complicated process, and there's a lot of data, i.e., tens of millions of scenes globally across the various Landsat satellites, all requiring extensive calibration, testing and validation.

DE Australia today produces ARD for Landsat and Sentinel-2 as well as a large range of derived data products, such as DEA Coastlines, which are embedded in decision making for the Australian Government. In 2019, DE Australia started moving to cloud computing for some ARD production, and later derived data automation. As of 2023, most DE Australia data is available directly from the cloud in cloud native formats accompanied by STAC metadata [8].

3.2. Digital Earth Africa

The Digital Earth Africa (DE Africa) program was established in 2019 [9] and initially ran in parallel to DE Australia in GA before being transitioned to an Africa-led host organization in 2022. While the DE Africa platform is fundamentally the same as DE Australia in terms of architecture, one key difference is that due to recent availability of high-quality global CEOS-ARD for Landsat and Sentinel-2, this program chose not to develop their own ARD data processing pipelines and instead takes a copy of existing collections. DE Africa also invested in a Sentinel-1 ARD pipeline, which is run by Sinergise, and delivers radiometrically terrain corrected synthetic aperture radar data over all of Africa [10].

The derived data processes, product development and collaborative science practices are similar between DE Australia and DE Africa, and DE Africa shares all its ARD and derived products as open access, cloud native data, see [11]. But the use of CEOS-ARD that is produced by other agencies means that a significant amount of time is freed up to deliver capacity development and new derived-data products.

3.3. Digital Earth Pacific

Digital Earth Pacific (DE Pacific) was launched in 2023 with broad goals around increasing access to Earth observation data across the Pacific [12]. As CEOS-ARD is now available in cloud native data formats on multiple cloud providers, DE Pacific doesn't take a copy of foundational ARD and simply uses that data in place.

The DE Pacific team and the broader community working on the platform can focus on delivering new products without spending time on ARD production, and without spending time on data synchronization, which requires technical complexity. The time and resources saved by DE Pacific and other organizations globally represents the value of these developments in global ARD availability. It significantly reduces the barrier to entry for organizations, large and small.

6. DISCUSSION

The availability of global certified CEOS-ARD data in cloud native formats alongside other global data products, such as ESA's global land cover product, WorldCover, each with a standardized STAC metadata enables incredible analyses with simple tools.

As an example, an individual can now run a workflow that:

- Identifies all the Landsat scenes over a year covering a region of interest
- Merges scenes together and masks out clouds
- Creates an annual summary modified normalized-difference wetness index band ratio product
- Masks out ocean using the ESA WorldCover data
- Identifies the coastline and draws a contour line across it.

This workflow may sound obvious and simple but being able to do it without storing any data locally and without writing any custom software, aside from the actual analysis steps, is unprecedented. The ability to conduct this analysis on a consumer laptop with consumer-level network connection increases the number of people who can undertake this work significantly. It's no longer limited to those who have access to a supercomputer or data centre.

Contrast this process with what DE Australia needed to do to get to the same point, where they needed to copy raw data from tapes, process this data through Level 1 and Level 2, develop software to index scenes and then use the index to load data. It is the difference between ten years and ten minutes.

The workflow described above can be driven by metadata from standard STAC APIs as well as cloud native data that are both available under an open license and openly accessible across the Internet. More and more people are embracing this cloud native geospatial way of working. which means that software or data science notebooks can be

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written that are far less dependent on the environment they are running in. Together, this means that work on Earth observation data is easier to share, more easily reproducible and it's easier to undertake collaborative research.

6. CONCLUSIONS

The emergence of cloud native geospatial in combination with the broad availability of CEOS-ARD means we're now able to do more with less. This is the culmination of decades of research and development in ARD, software, cloud infrastructure and standards, and while we shouldn't take this work for granted, we should ensure that we're making the most of the opportunity that use of the new cloud native geospatial paradigm represents.

We're able to now develop on consumer hardware without needing access to bespoke compute environments and we can share code more easily. The interoperability enabled by STAC and cloud native geospatial data means that mixing data from different collections is simple. And together, we can build better data products, and "*turn a flood of raw data into understandable information about our society and out planet.*" [1]

This is a great time to join the cloud native geospatial community to help realize the Digital Earth vision together.

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